



Interpretable generalized additive neural networks

Mathias Kraus^{a,*}, Daniel Tschernutter^b, Sven Weinzierl^a, Patrick Zschech^a

^a FAU Erlangen-Nürnberg, Lange Gasse 20, Nürnberg 90403, Germany

^b ETH Zurich, Weinbergstr. 56/58, Zurich 8092, Switzerland

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ABSTRACT

We propose Interpretable Generalized Additive Neural Networks (IGANN), a novel machine learning model that uses gradient boosting and tailored neural networks to obtain high predictive performance while being interpretable to humans. We derive an efficient training algorithm based on the theory of extreme learning machines, that allows reducing the training process to solving a sequence of regularized linear regressions. We analyze the algorithm theoretically, provide insights into the rate of change of so-called shape functions, and show that the computational complexity of the training process scales linearly with the number of samples in the training dataset. We implement IGANN in PyTorch, which allows the model to be trained on graphics processing units (GPUs) to speed up training. We demonstrate favorable results in a variety of numerical experiments and showcase IGANN's value in three real-world case studies for productivity prediction, credit scoring, and criminal recidivism prediction.

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1. Introduction

Machine learning models have found their way into various domains within the discipline of operations research, ranging from sales (Chou et al., 2022) and manufacturing (Senoner et al., 2022) to customer relationship management (Coussement et al., 2010; De Caigny et al., 2018) and supply chain management (Carbonneau et al., 2008). Especially, advanced models based on complex model structures, such as deep neural networks (Borchert et al., 2023) and boosted decision trees (Senoner et al., 2022), have become the focus of academic and industrial efforts over the past decade as they are known to be capable of capturing non-trivial patterns from problem-specific training data to generating recommendations with high prediction qualities (Janiesch et al., 2021; Kraus et al., 2020).

The internal structure of such complex models often renders them black boxes for decision-makers and other stakeholders. That is, it is not directly observable what information in the input data drives the models to generate their decisions, which is why they are hard to comprehend (Miller, 2019). However, from an organizational perspective, using transparent decision logic is often crucial, especially in high-stakes decision scenarios such as healthcare, justice, and finance, not only because of increased regulatory scrutiny

but also to gain trust in the model's output (Rudin, 2019; Rudin & Radin, 2019). Here, decision-makers are increasingly emphasizing the importance of algorithmic decision-making that aligns with business logic and intuition (Coussement & Benoit, 2021). In parallel, ethical considerations such as the fairness of analytics have driven debate and policymaking (Martens, 2022). The integration of legal and ethical requirements in operations research is becoming increasingly important, as evidenced by a survey by Rao & Greenstein (2022), which found that the vast majority of decision-makers plan to invest in responsible AI in the future.

In response to these challenges, we propose *Interpretable Generalized Additive Neural Networks* (IGANN), a novel intrinsically interpretable machine learning model offering high prediction qualities while remaining transparent and comprehensible to humans. IGANN belongs to the family of generalized additive models (GAMs) (De Bock et al., 2010; Hastie & Tibshirani, 1986). In GAMs, the input features are mapped independently of each other in a non-linear way, and the mappings are summed up afterward. These mappings are commonly known as *shape functions* (Lou et al., 2012). As they can take arbitrary forms, it is possible to capture complex relationships between individual features and the target.

In contrast to previous GAMs, which are either based on splines (e.g., Hastie & Tibshirani, 1986), trees (e.g., Lou et al., 2012; 2013) or deep neural networks (e.g., Agarwal et al., 2021; Yang et al., 2021a; 2021b), IGANN combines the ideas of gradient boosting and tailored sparsified neural networks to obtain high predictive performance. Thereby, IGANN offers several key advantages over

* Corresponding author.

E-mail addresses: mathias.kraus@fau.de (M. Kraus), dtschernutter@ethz.ch (D. Tschernutter), sven.weinzierl@fau.de (S. Weinzierl), patrick.zschech@fau.de (P. Zschech).

previous GAM approaches for machine learning practice. First, IGANN initializes the shape functions linearly and then gradually adapts to potential non-linearities. That is, IGANN is based on the concepts of linear modeling and only if necessary resorts to non-linearities. Second, using a scale parameter in IGANN, the rate of change of the shape functions can be directly controlled. Finally, IGANN achieves linear time complexity as it merely solves a sequence of regularized linear regressions.

With our work, we contribute to machine learning theory and practice in the following ways:

1. We propose a novel machine learning model from the family of generalized additive models using sparsified neural networks.
2. We derive an efficient training algorithm based on the theory of extreme learning machines, that allows reducing the training process to solving a sequence of regularized linear regressions.
3. We theoretically analyze the relationship between a scale parameter in IGANN and the rate of change of the shape functions.
4. We provide evaluation results of a large-scale experiment comparing state-of-the-art interpretable machine learning models across 114 datasets.
5. We showcase the value of IGANN on three real-world case studies from the fields of productivity prediction, credit scoring, and criminal recidivism prediction.
6. We implement IGANN using PyTorch, which allows IGANN to scale to huge datasets. The implementation is publicly available and free to use for academia and industry.¹

The remainder of the paper is structured as follows. [Section 2](#) discusses the conceptual background and previous research on interpretable machine learning. In [Section 3](#), the model architecture and the training process are described, and IGANN is analyzed from a theoretical perspective. In [Section 4](#), the proposed model is evaluated and compared against baseline models. [Section 5](#) demonstrates the use of IGANN in three real-world applications. Finally, [Section 6](#) discusses the implications of our work for research and practice, and [Section 7](#) concludes this study.

2. Background and related work

2.1. Model interpretability

With the widespread use of machine learning models, the need for model interpretability has strongly increased, especially in critical application domains such as credit scoring and criminal justice (Rudin & Radin, 2019). In general, model interpretability refers to a model's ability to explain or present its behavior in human-understandable terms (Doshi-Velez & Kim, 2017). Following this notion, interpretability can take several forms, as different stakeholders may have different expectations and requirements for interpretability depending on the specific use case. Likewise, the level of interpretability that is sufficient for one use case may not be sufficient for another (Rudin, 2019). To this end, Doshi-Velez & Kim (2017) distinguish between *application-grounded*, *human-grounded*, and *functionally grounded interpretability*. While the first two forms involve user-centric evaluation criteria, such as perceived complexity or required cognitive effort, the latter refers to a formal definition of interpretability and thus allows an objective assessment.

In this paper, we focus explicitly on functionally grounded model interpretability and consider the extent to which a model is able to reveal how model inputs (i.e., features) are processed in order to produce a predicted outcome (i.e., target). Re-

lated to this view, there are three dimensions that are commonly used to provide insights into a model's functional behavior: *global interpretability*, *local interpretability*, and *feature contribution* (Barredo Arrieta et al., 2020; Du et al., 2019). They are not mutually exclusive, but allow for evaluation from different perspectives. [Table 1](#) provides a brief description of the different dimensions, which are also used in the following sections to characterize IGANN's interpretability.

Apart from this, a further distinction can be made between *post-hoc interpretability* and *intrinsic interpretability* (Du et al., 2019; Rudin, 2019). Post-hoc interpretability requires the construction of a second model to explain the behavior of an existing model that usually exhibits a high degree of complexity.² This is in stark contrast to intrinsic interpretability, which is achieved by constructing models that are interpretable by design. Such models provide not just an approximate explanation of a model's functional behavior, but an *exact* description of how a model computes a prediction. In this work, we follow the paradigm of intrinsic interpretability by proposing a model that directly incorporates interpretable model structures.

2.2. Interpretability in operations research

The field of interpretable machine learning has a long history, dating back to the origins of linear models in the 19th century (Molnar et al., 2020) and the development of decision trees in the 1980s (Ciocan & Mišić, 2022). In recent years, however, the field has regained momentum, which is reflected in the growing interest of researchers within the discipline of operations research. In this context, the interpretability of machine learning models is particularly important to ensure that the decisions made by models can be easily understood and justified.

For example, in supply chain management, machine learning models are used to predict demand, optimize inventory levels, and improve logistics. The decisions made by these models can have significant impacts on the business and therefore require a careful assessment. Senoner et al. (2022) explore the application of machine learning to electronics supply chains. Focusing on process quality improvement in semiconductor manufacturing, the authors develop a novel data-driven decision model. The authors propose a model that incorporates SHAP to analytically explain how system parameters and manufacturing process quality are correlated. This helps to optimize the manufacturing process in terms of quality.

Similarly, interpretable models have also been applied in the domain of customer relationship management. For example, Mitrović et al. (2018) examined the retention of features and feature types for prepaid and postpaid churn prediction models, providing insights into which factors are important for predicting customer churn behavior. De Caigny et al. (2018) proposed a hybrid segmented-modeling approach that combines logistic regression and decision trees to explain the main churn drivers for different customer segments, thereby unraveling the black box for marketing managers. Interpretable models can also help companies ensure compliance with regulations and ethical considerations. For instance, the use of black box models in credit risk management may lead to unfair lending decisions and hide unknown biases, which can be addressed by using interpretable models that provide insights into the determinants of credit losses (Bastos & Matos, 2022). In addition, some researchers have used generalized additive models to study customer behavior, such as the identification of drivers for customer loyalty and churn

² Common techniques aiming at post-hoc interpretability are Shapley additive explanations (SHAP) (Lundberg et al., 2020) and local interpretable model agnostic explanations (LIME) (Ribeiro et al., 2016).

¹ <https://github.com/MathiasKraus/igann>

Table 1
Three dimensions of functional interpretability of ML models.

Dimension	Description	Examples
Global interpretability	Describes the ability to understand the overall decision-making process of a model. This type of interpretability is useful for understanding the model's behavior as a whole and for identifying any potential biases or issues.	Linear models/ generalized additive models (global functional structure, i.e., $f(\cdot)$), decision trees (global tree structure)
Local interpretability	Refers to the ability to understand the reasoning behind an individual model prediction, or why a particular prediction was made for a specific input. This type of interpretability is particularly useful for situations where the model's decision-making process must be transparent to stakeholders who are affected by a prediction.	Linear models/ generalized additive models (evaluated on a given point, i.e., $f(x_0)$), decision trees (individual decision path)
Feature contribution	Provides insights into the relative importance of each feature in the model's decision-making process. These methods can help identify which features are most important in determining the model's output, which can be particularly useful for feature selection and engineering.	Linear regression coefficients, Permutation feature importance

(Coussement et al., 2010), or the prediction of customer lifetime value and churn (Coussement et al., 2015).

Healthcare is another critical domain where the interpretability of machine learning models is usually a strict requirement for the models to be applied in practice (Rudin, 2019). As such, previous research has proposed a hybrid methodology for feature selection using genetic algorithms to predict and explain the quality of life for lung transplant patients (Al-Ebbini et al., 2017). Another study proposed an analytics methodology that developed three different types of prediction models to predict and explain the severity of inflammation in patients with Crohn's disease (Reddy et al., 2019). Additionally, researchers have developed a clinical decision support system for predicting diabetes, and identified the most important variables for detecting diabetes through their feature contribution (Piri et al., 2017).

2.3. Generalized additive models

In this paper, we focus on generalized additive models as a specific type of interpretable models. They have been widely used in high-stake domains where model transparency is a pivotal requirement (Barredo Arrieta et al., 2020; Zschech et al., 2022). In general, GAMs refer to a type of machine learning models which build dependencies between input features and the target by adding several individual non-linear mappings, called *shape functions*. Input features are constrained to be independent of each other, which allows the assessment of the learned shape functions after training. Formally, GAMs are defined as

$$f(x) = f_1(x_1) + f_2(x_2) + \dots + f_m(x_m), \tag{1}$$

where each f_i describes a shape function that maps the i th input feature to the output space.

The original GAM refers to a model in which the shape functions represent regression splines with a flexible choice of orders to partially control the smoothness³ of the resulting shape functions (Hastie & Tibshirani, 1986). De Bock et al. (2010) extended this approach by integrating GAMs as base classifiers into an ensemble learning strategy. The authors proposed three alternative strategies based on bagging, a random subspace method, and a combination of both. However, spline-based models are often considered too smooth, and they have shown inferior prediction qualities as compared to more complex models, such as tree-based GAMs (Caruana et al., 2015; Lou et al., 2012).

By using tree-based GAMs, Lou et al. (2012) demonstrated on real-world datasets that much better prediction results can be achieved by using bagged and boosted tree ensembles for fitting the shape functions. The authors further enhanced this tree-

based approach (e.g., by considering pairwise interaction effects Lou et al., 2013), and made it publicly available as an easy-to-use algorithm, known as *Explainable Boosting Machine* (EBM) (Nori et al., 2019). EBM has the advantage of being very fast and sufficiently flexible to capture the complex dependencies in real-world datasets, such as sharp jumps due to abrupt changes or discrete thresholds (Chang et al., 2021; Lou et al., 2012). On the downside, the approach carries the risk of producing unexpected discontinuities that are difficult to justify. This effect may become even worse in the presence of noisy samples and outliers.

More recently, the focus has shifted towards the development of GAM extensions inspired by artificial neural networks. For example, Agarwal et al. (2021) proposed *Neural Additive Models* (NAM) in which feature-wise shape functions are learned via deep neural networks (DNNs) consisting of multiple hidden layers. To fit the rapid changes in the target variables of real-world data, the authors introduce exp-centered hidden units. These units allow to capture sharp changes in the output accurately. However, this approach tends to overfit the training samples heavily and therefore requires an extensive hyperparameter search. Likewise, the model is computationally expensive as it requires the training of multiple DNNs.

Another DNN-based approach was pursued by Yang et al. (2021b). The authors proposed *Generalized Additive Models with Structured Interactions* (GAMI-Net). The basic architecture is similar to that of a NAM, using a simple GAM-based structure in the form of a disentangled feed-forward network with multiple additive sub-networks. However, to reduce unnecessary model complexity and avoid overfitting, GAMI-Net integrates sparsity constraints to select the most relevant features for a parsimonious representation. In addition, GAMI-Net has the advantage of being able to capture the pairwise interactions between the individual features, which can lead to better predictive performance. However, the detection of the relevant interaction terms is computationally expensive, and in the case of GAMI-Net, requires further model constraints in the form of heredity and marginal clarity constraints to enhance structural interpretability and avoid mutual absorption between the primary effects and pairwise interactions. Moreover, in analogy to NAM, GAMI-Net suffers from long training times as it requires training multiple computationally expensive deep sub-networks.

A different architectural design was sought with the proposal of *Explainable Neural Network* (xNN) (Vaughan et al., 2018) and its successor *Enhanced Explainable Neural Network* (ExNN) (Yang et al., 2021a). Similar to GAMI-Net, both models are based on multiple sub-networks and regularization terms to retain sparsity and receive a parsimonious model. However, instead of using a simple GAM structure, both models are based on the structure of additive index models by adopting the idea of a projection pursuit regression (Friedman & Stuetzle, 1981). This structure generally violates the principle of univariate feature mappings due to an

³ Note that related literature is followed in the field of interpretable AI and the term "smoothness" is used to describe the rate of change qualitatively, i.e., not in the strict mathematical sense.

additional projection layer that fully connects all input features to the following sub-networks so that each feature can possibly have a partial contribution to all corresponding shape functions. Although this design can lead to higher predictive performance when fitting mathematically complex functions, the model output is hardly interpretable in a meaningful way for real-world datasets, especially when several features are involved (Zschech et al., 2022).

To sum up, recent advancements show several improvements over the original GAM approach which have steadily enhanced the field in terms of predictive performance and/or interpretability. However, there are several limitations, for instance, non-intuitive shape functions due to stepwise modeling, limited control over the rate of change of shape functions, long training times, or high sensitivity to hyperparameter settings. Therefore, we aim to address these limitations by proposing IGANN in the next section.

3. Interpretable generalized additive neural networks (IGANN)

In this section, we propose Interpretable Generalized Additive Neural Networks (IGANN). First, Section 3.1 briefly discusses the GAM framework in the current context and introduces our novel shape functions based on single hidden layer neural networks. Second, Section 3.2 analyzes the resulting learning task theoretically and proposes a highly scalable training algorithm. Finally, the theoretical properties of IGANN are derived and the computational complexity of the training algorithm is discussed in Section 3.3.

3.1. Architecture of IGANN

As motivated, we propose to initialize our model with linear shape functions and gradually incorporate more complex dynamics by adding non-linear base functions from a tailored function class $\mathcal{F}_{\text{SNN}}$. While the resulting GAM is formally defined in Section 3.1.1, we discuss the class of base functions $\mathcal{F}_{\text{SNN}}$ in detail in Section 3.1.2.

3.1.1. Generalized additive model framework

Mathematically, the model predictions can be defined as follows. Let the training data be given as $\{(x^i, y^i)\}_{i=1}^n$, where n is the number of training samples, $x^i \in \mathbb{R}^d$ with d being the dimension of the feature space representing the covariates, and y^i the targets. For regression tasks, we assume $y^i \in \mathbb{R}$, whereas binary classification tasks require $y^i \in \{-1, 1\}$. The model prediction $\hat{y}$ for $x \in \mathbb{R}^d$ is then defined as

$$\hat{y} = \underbrace{\langle a, x \rangle + b}_{\text{linear component}} + \underbrace{\sum_{l=1}^L s_l f_l(x)}_{\text{non-linear base functions}}, \quad (2)$$

where $a \in \mathbb{R}^d$ are the parameters of the linear component, $b \in \mathbb{R}$ is the intercept, $f_l \in \mathcal{F}_{\text{SNN}}$ are the base functions, $s_l > 0$ is the l th step size, and $L \in \mathbb{N}$ is the number of base functions in the GAM. Note that for binary classification tasks, we are working in logit space, i.e., the actual predicted class probability is given by $\phi(\hat{y})$, where ϕ is the logistic function.

3.1.2. Structure of shape functions

One of the primary contributions is the tailored form of the base functions in $\mathcal{F}_{\text{SNN}}$. The latter are based on single hidden layer neural networks that directly induce the form of the proposed shape functions. The idea is to remove feature interactions in the first layer by subdividing the hidden layer into d components (one for each feature dimension), each with a predefined number of hidden neurons $N \in \mathbb{N}$. This can be done by sparsifying the weights in the first layer. Mathematically, the base functions are defined

as follows. Let σ be a non-constant piecewise continuous activation function (e.g., ReLU, ELU). The weights of the k th component of the first layer are denoted with $w^k \in \mathbb{R}^N$ and the output layer by $\alpha^k \in \mathbb{R}^N$ for $k \in \{1, \dots, d\}$. The base function with parameters $\theta = (\alpha^1, \dots, \alpha^d, w^1, \dots, w^d)$ is then defined by

$$f_\theta(x) = \sum_{k=1}^d \sum_{j=1}^N \alpha_j^k \sigma(x_k w_j^k), \quad (3)$$

where subscripts denote the entries of the given vectors, e.g., x_k denotes the k th entry of x . Again, note that the proposed base functions can be seen as sparsified single hidden layer neural networks, i.e., for each $\theta \in \mathbb{R}^{2dN}$, there exist $\tilde{\alpha} \in \mathbb{R}^{N^2}$ and $\tilde{w}^j \in \mathbb{R}^{N^2}$ for $j \in \{1, \dots, N^2\}$, such that

$$f_\theta(x) = \sum_{j=1}^{N^2} \tilde{\alpha}_j \sigma(\langle x, \tilde{w}^j \rangle). \quad (4)$$

Fig. 1 illustrates this idea for three input features and four hidden neurons. Finally, the class of base functions is defined as

$$\mathcal{F}_{\text{SNN}} = \{f_\theta : \theta \in \mathbb{R}^{2dN}\}. \quad (5)$$

3.2. Scalable training of IGANN

The idea behind the proposed training algorithm builds on the theory of so-called gradient boosting. That is, the parameters a and b of the linear component are estimated using standard methods, i.e., a Lasso model (least absolute shrinkage and selection operator) with regularization parameter λ^{Reg} for regression tasks or a logistic regression model with regularization parameter λ^{Class} for binary classification tasks. Then, the base functions are successively added until a certain stopping criterion is met or a maximal number of base functions is reached, see Section 3.2.1. Adding additional base functions involves the solution of non-trivial optimization problems. Hence, we propose an efficient training algorithm based on the theory of extreme learning machines in Section 3.2.2. A pseudocode of the proposed algorithm is provided in Section 3.2.3.

3.2.1. Gradient boosting

The training task for IGANN consists of minimizing the following regularized loss function

$$\mathcal{L}(a, b, \theta_1, \dots, \theta_L) = \frac{1}{n} \sum_{i=1}^n l_{\text{task}}(\hat{y}^i, y^i) + \lambda^{\text{task}} \|a\|_1 + \frac{\lambda^{\text{GB}}}{n} \sum_{l=1}^L \|\theta_l\|^2, \quad (6)$$

where $\text{task} \in \{\text{Reg}, \text{Class}\}$, λ^{GB} is a regularization parameter, and θ_l is the parameter vector of the l th base function. For regression, the squared loss is used, whereas for binary classification tasks, the logistic loss is used, i.e.,

$$l_{\text{Reg}}(\hat{y}, y) = (\hat{y} - y)^2, \quad (7)$$

$$l_{\text{Class}}(\hat{y}, y) = \log(1 + e^{-y\hat{y}}). \quad (8)$$

The idea of gradient boosting is now to train the model in an additive manner (see, e.g., Chen & Guestrin, 2016). That is, if $\hat{y}^{i,t-1}$ gives the prediction of the model in stage $t-1 \in \{0, \dots, L\}$, i.e.,

$$\hat{y}^{i,t-1} = \langle a, x^i \rangle + b + \sum_{l=1}^{t-1} s_l f_{\theta_l}(x^i), \quad (9)$$

we add f_{θ_t} to minimize

$$\mathcal{L}^{(t)}(\theta_t) = \frac{1}{n} \sum_{i=1}^n l_{\text{task}}(\hat{y}^{i,t-1} + s_t f_{\theta_t}(x^i), y^i) + \frac{\lambda^{\text{GB}}}{n} \|\theta_t\|^2. \quad (10)$$

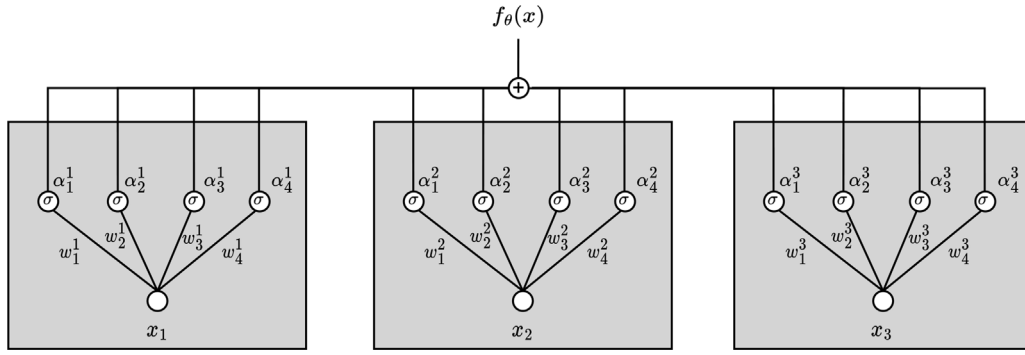


Fig. 1. Illustration of the proposed base function for a problem with $d = 3$ input features and $N = 4$ hidden neurons. The model represents a single hidden layer neural network, in which feature interactions are eliminated by restricting the weights of the input layer to individual features, as indicated by gray boxes.

In this manner, new basis functions that improve the model predictions are added according to Eq. (6). Note that in stage $t = 0$, the corresponding regression problem is simply solved without any base functions

$$\frac{1}{n} \sum_{i=1}^n l_{\text{task}}(\hat{y}^i, y^i) + \lambda^{\text{task}} \|a\|_1, \quad (11)$$

i.e., we either fit a lasso model or a logistic regression with ℓ_1 -penalty, and then set $\hat{y}^{i,0} = \langle a, x^i \rangle + b$.

For computational purposes, gradient boosting techniques often make use of second-order Taylor approximations of the loss function in stage t (see, e.g., Badirli et al., 2020; Chen & Guestrin, 2016), i.e., Eq. (10). That is, instead of solving $\min_{\theta_t} \mathcal{L}^{(t)}(\theta_t)$ accurately, one solves the approximation

$$\begin{aligned} \min_{\theta_t} \frac{1}{n} \sum_{i=1}^n \left[l_{\text{task}}(\hat{y}^{i,t-1}, y^i) + s_t \frac{\partial}{\partial \hat{y}} l_{\text{task}}(\hat{y}^{i,t-1}, y^i) f_{\theta_t}(x^i) \right. \\ \left. + \frac{s_t^2}{2} \frac{\partial^2}{\partial \hat{y}^2} l_{\text{task}}(\hat{y}^{i,t-1}, y^i) f_{\theta_t}^2(x^i) \right] \\ + \frac{\lambda^{\text{GB}}}{n} \|\theta_t\|^2, \quad \text{SOA}_t \end{aligned}$$

Note that the terms $f_{\theta_t}(x^i)$ are now outside l_{task} .

3.2.2. Efficient boosting steps

A crucial part of gradient boosting algorithms is the successive solution of (SOA_t) in an efficient manner. To ensure scalability, the proposed algorithm builds upon the theory of extreme learning machines (Huang et al., 2006). That is, our idea is to merely sample the weights in the first layer of the base functions according to a fixed distribution and then solve (SOA_t) solely with respect to the weights in the hidden layer. Mathematically, we proceed as follows. Let $\theta_t = (\alpha^{1,t}, \dots, \alpha^{d,t}, w^{1,t}, \dots, w^{d,t})$ and sample the weights of the first layer according to

$$w^{k,t} \sim \mathcal{N}(\vec{0}, \gamma I), \quad (12)$$

where $\vec{0} \in \mathbb{R}^d$ is the zero vector, $I \in \mathbb{R}^{d \times d}$ is the identity matrix, and γ is a scale parameter. Now, besides our novel framework that combines the theories of gradient boosting and extreme learning machines, our primary result with regard to our optimization algorithm is that (SOA_t) reduces to a regularized linear regression problem, if the weights $(w^{1,t}, \dots, w^{d,t})$ are fixed and one optimizes solely with respect to $(\alpha^{1,t}, \dots, \alpha^{d,t})$. The details are summarized in Proposition 1.

Proposition 1. For fixed weights $(w^{1,t}, \dots, w^{d,t})$ in stage t , (SOA_t) reduces to the regularized linear least squares problem

$$\min_{\alpha_t = (\alpha^{1,t}, \dots, \alpha^{d,t})} \|V^t \alpha_t - \tilde{y}^t\|^2 + \lambda^{\text{GB}} \|\alpha_t\|^2, \quad \text{LLS}_t$$

with $V^t \in \mathbb{R}^{n \times dN}$ and $\tilde{y}^t \in \mathbb{R}^n$ solely depending on the current predictions $\{\hat{y}^{i,t-1}\}_{i=1}^n$ in stage t and the step size – or boosting rate – $s_t > 0$.

Proof. See Appendix A $\square$

3.2.3. Pseudocode

Proposition 1 allows for a very efficient training algorithm that is summarized in Algorithm 1. For ease of notation, the step size – or boosting rate – is set to $s_t = s > 0$ for all $t \geq 1$. Note that the boosting rate can be set differently in every stage. That is, our theory is flexible enough to incorporate adaptive boosting rate approaches in future work, which can often lead to better performance (cf. Schaul et al., 2013; Xu et al., 2019; Zeiler, 2012).

Algorithm 1: Interpretable generalized additive neural networks (IGANN).

Input: Number of hidden neurons N , boosting rate s , scale parameter γ , maximal number of iterations $\mathcal{T}$
Output: Model parameters $(a, b, \theta_1, \dots, \theta_{t-1})$
 Initialize IGANN with a linear model $\langle a, x \rangle + b$ by solving problem (11);
 Set $t \leftarrow 1$;
while stopping criterion not met **and** $t < \mathcal{T}$ **do**
 Update $\hat{y}^{i,t-1}$ according to Equation (9);
 Sample the weights of the input layer $w^{k,t} \sim \mathcal{N}(\vec{0}, \gamma I)$ for $k \in \{1, \dots, d\}$;
 Compute V^t and $\tilde{y}^t$ according to Proposition 1 with $s_t = s$;
 Compute the weights of the hidden layer α_t by solving the regularized linear least squares problem (LLS_t);
 Set $t \leftarrow t + 1$;
end
return $(a, b, \theta_1, \dots, \theta_{t-1})$;

In our implementation, early stopping is used in Algorithm 1.

3.3. Theoretical analysis of IGANN

In the following, we derive the theoretical properties of IGANN. According to the proposed additive model framework (see Eq. (2)), the shape functions are in the form

$$S_k(x_k) = a_k x_k + s \sum_{l=1}^L \sum_{j=1}^N \alpha_j^{k,l} \sigma(x_k w_j^{k,l}) \quad (13)$$

for $k \in \{1, \dots, d\}$. Fig. 2 illustrates the structure of these functions exemplary for three input features, four hidden neurons, and three base functions.

Following the formal definition of GAMs in Eq. (1), the output of IGANN is thus given in the form

$$\hat{y} = b + S_1(x_1) + S_2(x_2) + \dots + S_d(x_d). \quad (14)$$

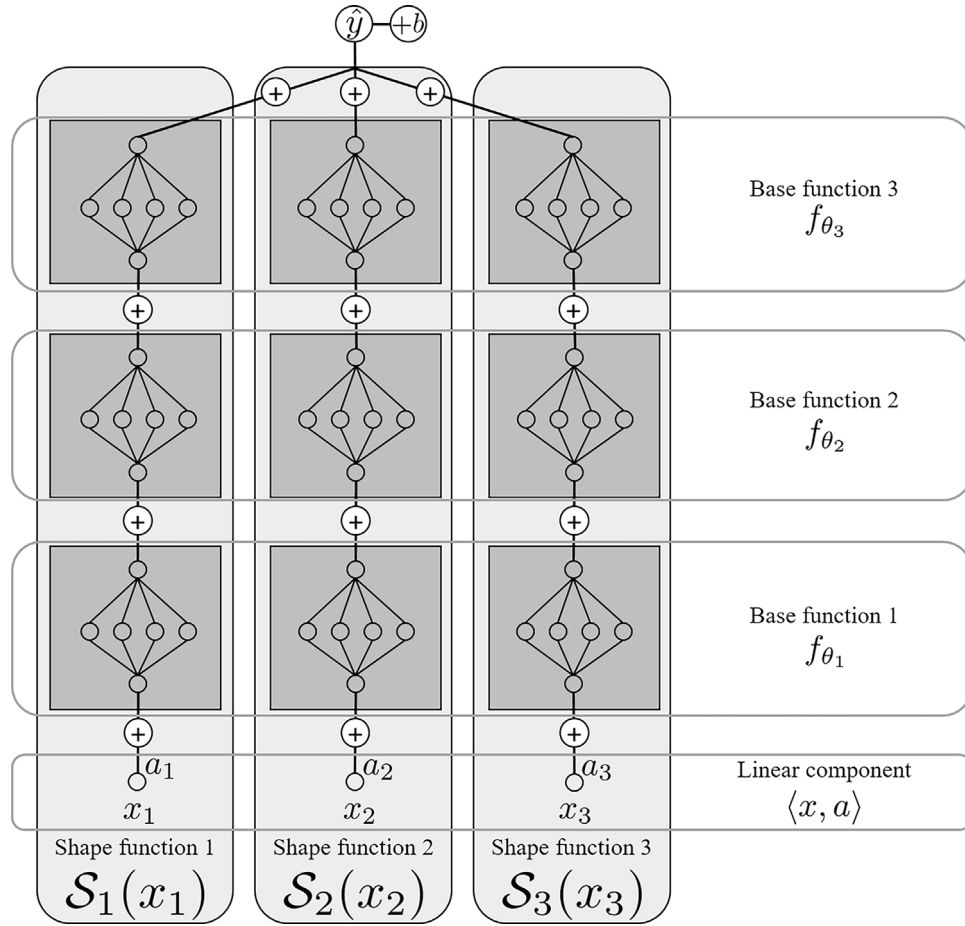


Fig. 2. Illustration of the structure of derived shape functions.

Section 3.3.1 analyzes the rate of change of these functions and derives connections to the scaling parameter γ . Second, Section 3.3.2 establishes the computational complexity of Algorithm 1.

3.3.1. Rate of change

The rate of change of the shape functions is an important aspect when it comes to interpretability as it formalizes the slope of the function and, hence, the sensitivity of changes in the covariates. Proposition 2 shows that the rate of change of the shape functions can be uniformly bounded from above.

Proposition 2. For a Lipschitz continuous activation function σ , the rate of change of the k th shape function S_k , i.e.,

$$\sup_{x, y \in \mathbb{R}} \frac{|S_k(x) - S_k(y)|}{|x - y|}, \quad (15)$$

can be bounded by

$$|a_k| + C \left(\max_{l=1, \dots, L} \|\tilde{y}^l\|_\infty \right) w_\infty^2, \quad (16)$$

with a constant $C > 0$ and $w_\infty = \max_{k, l, j} |w_j^{k, l}|$.

Assuming that $\|\tilde{y}^l\|_\infty$ stays bounded from above, we thus have an upper bound for the rate of change of S_k in the form

$$|\alpha_k| + \tilde{C} w_\infty^2, \quad (17)$$

where $\tilde{C} > 0$ is a constant. The first summand describes the rate of change induced by the linear part and the second summand the

rate of change induced by the additional non-linearities. From the properties of the normal distribution, one obtains a direct relationship between the size of our scaling parameter γ and w_∞ and thus the upper bound for the rate of change. For empirical evidence, see our experiments in Appendix B.

3.3.2. Computational complexity

The computational complexity of Algorithm 1 is mainly driven by the involved linear regression subproblems and is thus given by $\mathcal{O}((T+1)(d^3 + d^2n))$. In a standard setting with $d \ll n$ and $T \ll n$, we thus yield a linear time complexity $\mathcal{O}(n)$. This is the result of our carefully crafted boosting steps, allowing IGANN to scale to large-scale problems with millions of samples and hundreds of features. Besides our theoretical derivations, we demonstrate the scalability of IGANN empirically in Appendix C. Here, also our tailored implementation for running our algorithm on GPUs results in consistent speedups of nearly an order of magnitude compared to calculations on CPUs.

3.3.3. Interpretability of IGANN

In the following, we discuss the three dimensions of functional interpretability discussed in Section 2.1 in the context of our model framework.

Global interpretability From the definition of shape functions in Eq. (13), we see that S_k depends solely on the k th component of the input vector $x \in \mathbb{R}^d$, i.e., $x_k \in \mathbb{R}$. Hence, each shape function takes a one-dimensional input and returns a one-dimensional output and can thus be visualized well (see example in Fig. 3). The x-axis corresponds to the values of the input feature x_k , whereas the y-axis corresponds to the value of the shape function $S_k(x_k)$. The

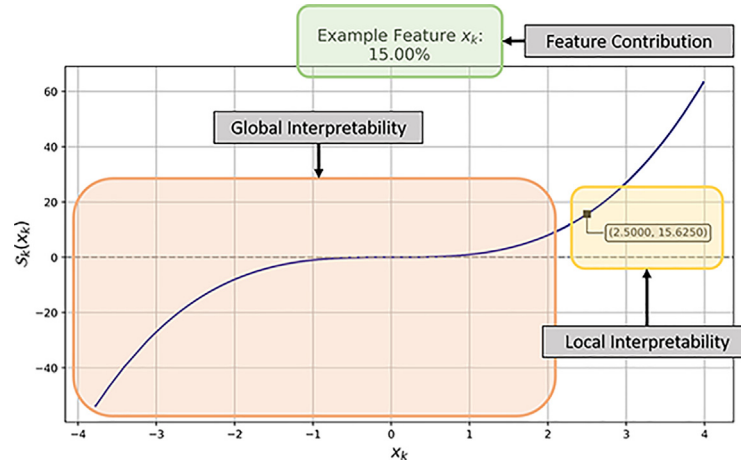


Fig. 3. Exemplary visualization of shape function.

line plot depicts the effect of the corresponding feature value on the predicted output. Thus, the overall decision logic of our model can be made intelligible by providing shape plots of the input features.

Local interpretability The final prediction of our model is given by the sum of all shape functions and the intercept (cf. Eq. (14)). That is, individual predictions can be decomposed into their respective feature components and thus yield insights into the decision logic on a local level. Thus, we can shift from a global point of view $S_k(\cdot)$ to the evaluation on a specific input x_k^0 , i.e., $S_k(x_k^0)$. In Fig. 3, for instance, the value of the shape function for $x_k^0 = 2.5$ is given by $S_k(2.5) = 15.625$. In the context of Eq. (14), the additive structure

$$\hat{y}^0 = b + S_1(x_1^0) + S_2(x_2^0) + \dots + \underbrace{S_k(2.5)}_{=15.625} + \dots + S_d(x_d^0), \quad (18)$$

thus yields additional insights into the corresponding model output at x^0 and the corresponding effect of feature k in this context.

Feature contribution Furthermore, the share of each shape function in the output sum provides a natural measure of feature contribution on an aggregated level and thus a notion of feature importance given the training data. This percentage value is displayed in the title of Fig. 3 together with the corresponding feature name. Mathematically, the computation is as follows. For given training covariate vectors $x^1, \dots, x^n$, we iterate for each $k \in \{1, \dots, d\}$ through the unique training feature values and sum up the contribution that each base function has on the model output. After normalization, these values describe the average share that each shape function has on the model prediction. Details are summarized in Algorithm 2.

4. Performance evaluation of IGANN

Our experimental setup demonstrates the potential of IGANN as a general out-of-the-box machine learning model. In particular, we focus on two main characteristics in our evaluation: First, we evaluate and compare the predictive performance of IGANN with various state-of-the-art interpretable models across a manifold of datasets. Our extensive numerical experiments are based on 50 regression and 64 binary classification datasets from various domains, see Appendix D. Second, we evaluate IGANN and all baseline models once as a (hyperparameter-)tuned version and once as a default version (with standard hyperparameters).

Algorithm 2: Calculation of feature contribution.

Input: IGANN model parameters $a_k, \alpha_j^{k,l}, w_j^{k,l}$ for $j \in \{1, \dots, N\}, k \in \{1, \dots, d\}, l \in \{1, \dots, L\}$, covariate vectors $x^1, \dots, x^n$

Output: Feature contributions $\zeta_1, \dots, \zeta_d$

Set $\zeta_1, \dots, \zeta_d \leftarrow 0$;

/ Compute unique training values for features */*

for $k = 1, \dots, d$ **do**

 Compute $n_k \leq n$ unique training values for feature k from all training samples via $\tilde{x}_k^1, \dots, \tilde{x}_k^{n_k} \leftarrow \text{unique}(x_k^1, \dots, x_k^n)$;

end

/ Compute mean prediction of each shape function */*

for $k = 1, \dots, d$ **do**

for $i = 1, \dots, n_k$ **do**

 Set $\zeta_k \leftarrow a_k \tilde{x}_k^i$;

for $l = 1, \dots, L$ **do**

 Set $\zeta_k \leftarrow \zeta_k + s \sum_{j=1}^N \alpha_j^{k,l} \sigma(\tilde{x}_k^i w_j^{k,l})$;

end

end

 Set $\zeta_k \leftarrow \zeta_k / n_k$;

end

/ Compute feature contribution as relative contribution to the average IGANN model prediction */*

Compute mean model prediction $\Psi \leftarrow \sum_{k=1}^d \zeta_k$;

for $k = 1, \dots, d$ **do**

 Set $\zeta_k \leftarrow \zeta_k / \Psi$;

end

return $\zeta_1, \dots, \zeta_d$;

4.1. Baseline models

Our baseline models comprise state-of-the-art interpretable machine learning models with a sufficiently short computational time complexity to perform all training tasks within a reasonable time budget. We decided to compare IGANN with the linear models Lasso and Ridge, and the non-linear models decision tree, EBM, and GAM-Splines. We decided against neural network-based models as the training times are usually magnitudes higher and require expert knowledge to tune the hyperparameters, thus, preventing widespread usability and an interactive data analytics workflow (Zschech et al., 2022).

4.2. Preprocessing and evaluation strategy

We use minimal preprocessing to ensure a fair comparison between models. In fact, we only standardize the features (and tar-

Table 2

Model performances for default hyperparameters (top) and tuned hyperparameters (bottom) in terms of ranking, mean squared error (MSE), and wins/losses against IGANN across 50 regression datasets. Standard deviation in parentheses.

Model	Ranking	MSE	W/L
Lasso-default	5.92***	1.013 (0.024)	0/50
Ridge-default	2.98	0.386 (0.249)	20/30
Decision-Tree-default	3.98***	0.468 (0.357)	8/42
GAM-Splines-default	3.08	0.408 (0.257)	20/30
EBM-default	2.66	0.359 (0.222)	21/29
IGANN-default	2.38	0.345 (0.239)	–/–
Lasso-tuned	4.00***	0.379 (0.248)	8/42
Ridge-tuned	4.20***	0.382 (0.249)	6/44
Decision-Tree-tuned	3.82***	0.322 (0.223)	11/39
GAM-Splines-tuned	3.32***	0.333 (0.228)	11/39
EBM-tuned	3.66***	0.349 (0.220)	14/36
IGANN-tuned	2.00	0.314 (0.226)	–/–

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

gets in regression tasks) by removing the mean and scaling to unit variance. The scaling parameters are estimated on the training set.

Following common practice, we employ 5-fold cross-validation to assess the machine learning models.⁴ We follow Maldonado et al. (2022) and report the following metrics for each model: we calculate the average mean squared error (MSE) for regression and the log loss (Lloss) for classification tasks. Additionally, we report the average rank of the models, with 1 being the best rank. Based on the ranks, we apply the Friedman test to assess the predictive performance of IGANN compared to the baseline models (Demšar, 2006). The null hypothesis being tested is that there is no difference in performance between IGANN and the baseline models and, hence, that the ranks of their performance are equal. If the null hypothesis is rejected, the Holm post-hoc test is used to compare IGANN to the other algorithms in a pairwise manner (García et al., 2010). Finally, we report the number of wins and losses (W/L) for each method compared to IGANN, using the MSE or Lloss values.

To gain a comprehensive understanding of our model's performance, we perform additional evaluations on subsets of the datasets. These evaluations allow us to examine the model's ability to handle datasets with varying sizes and dimensionalities. We divide the datasets into two groups based on sample size, with the division line being set at the median number of samples. Similarly, we divide the datasets into two groups based on dimensionality, with the division line set at the median number of features. We list details on model implementations and the grid of hyperparameters in Appendix G. All models are trained on a workstation with an Intel i7-12700K with 12 cores and 128 gigabyte of memory.⁵

4.3. Performance results

In the following, we report the model performances on regression and classification tasks. We provide detailed results in Appendix E.

4.3.1. Regression tasks

Table 2 shows the performance of the machine learning models for default hyperparameters (top) and tuned hyperparameters (bottom) across 50 regression datasets. In both cases, IGANN outperforms all baseline models, showing the lowest average rank and the lowest mean squared error. While the improvements in terms

Table 3

Model performances for stratified regression datasets in terms of ranking, mean squared error (MSE), and wins/losses against IGANN. Standard deviation in parentheses.

Model	Ranking	MSE	W/L
(a) Small datasets			
Lasso-tuned	3.32***	0.296 (0.245)	7/18
Ridge-tuned	3.56***	0.300 (0.249)	6/19
Decision-Tree-tuned	3.88***	0.295 (0.220)	4/21
GAM-Splines-tuned	3.76***	0.304 (0.243)	3/22
EBM-tuned	4.56***	0.329 (0.232)	3/22
IGANN-tuned	1.92	0.266 (0.228)	–/–
(b) Large datasets			
Lasso-tuned	4.68***	0.463 (0.225)	1/24
Ridge-tuned	4.84***	0.464 (0.225)	0/25
Decision-Tree-tuned	3.76***	0.350 (0.227)	7/18
GAM-Splines-tuned	2.88	0.362 (0.214)	8/17
EBM-tuned	2.76	0.369 (0.211)	11/14
IGANN-tuned	2.08	0.362 (0.218)	–/–
(c) Low-dimensional			
Lasso-tuned	3.77***	0.399 (0.292)	2/20
Ridge-tuned	3.77***	0.402 (0.294)	2/20
Decision-Tree-tuned	4.14***	0.359 (0.230)	4/18
GAM-Splines-tuned	3.41***	0.375 (0.259)	5/17
EBM-tuned	4.09***	0.394 (0.254)	5/17
IGANN-tuned	1.82	0.354 (0.254)	–/–
(d) High-dimensional			
Lasso-tuned	4.18***	0.364 (0.211)	6/22
Ridge-tuned	4.54***	0.366 (0.212)	4/24
Decision-Tree-tuned	3.57**	0.293 (0.218)	7/21
GAM-Splines-tuned	3.25**	0.300 (0.200)	6/22
EBM-tuned	3.32**	0.314 (0.187)	9/19
IGANN-tuned	2.14	0.282 (0.200)	–/–

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

of the ranking are not statistically significant for default hyperparameters, they are significant on the 0.01 level for the tuned models.

In terms of training times for the tuned hyperparameters, we find that EBM requires on average 0.228 (0.455) seconds, GAM-splines 0.159 (0.357) seconds, and IGANN 0.539 (2.407) seconds, where the number in parentheses denotes the standard deviation. The linear models, as well as the decision tree, range between 0.001 and 0.024 seconds.

Table 3 shows the results of all tuned models for various subsets of the datasets. For small and low-dimensional datasets, we find significant improvements of IGANN over baseline models on the 0.01 significance level. On high-dimensional datasets, these improvements are still significant, yet on the 0.05 level. On large datasets, we find no significant improvements of IGANN over GAM-Splines and EBM. Nevertheless, the average ranking is still the lowest.

4.3.2. Classification tasks

Table 4 summarizes the model performances across 64 binary classification datasets for default hyperparameters (top) and tuned hyperparameters (bottom). Here, we find that EBM achieves the best average ranking in both cases with IGANN having the second-best ranking. Regarding the average log loss, IGANN outperforms all baseline models for default hyperparameters and comes third for tuned hyperparameters.

In terms of training times for the tuned hyperparameters, we find that EBM requires on average 0.366 (1.41) seconds, GAM-splines 2.132 (5.313) seconds, and IGANN 2.20 (10.05) seconds, where the numbers in parentheses denote the standard deviations.

Table 5 shows the classification results for various subsets of the 64 datasets for tuned hyperparameters. Again, the stratification is done by grouping the datasets into small, large, low- and high-dimensional datasets. Here, we find that IGANN shows the best rank on low-dimensional datasets. On large datasets, IGANN

⁴ Additionally, we set aside a hold-out set in the beginning and evaluate the model with the best cross-validation score on it. Results are reported in Appendix F.

⁵ We note that the memory consumption of all models in our experiments is negligible.

Table 4

Model performances for default hyperparameters (top) and tuned hyperparameters (bottom) in terms of ranking, log loss (LLoss), and wins/losses against IGANN across 64 binary classification datasets. Standard deviation in parentheses.

Model	Ranking	LLoss	W/L
Lasso-default	3.14	0.379 (0.197)	28/36
Ridge-default	3.44	0.388 (0.205)	28/36
Decision-Tree-default	5.53***	6.411 (5.548)	6/58
GAM-Splines-default	3.33	0.381 (0.213)	28/36
EBM-default	2.48	0.368 (0.221)	43/21
IGANN-default	3.08	0.363 (0.224)	—/—
Lasso-tuned	4.16***	0.373 (0.195)	14/50
Ridge-tuned	3.50	0.368 (0.194)	23/41
Decision-Tree-tuned	4.56***	0.518 (0.600)	18/46
GAM-Splines-tuned	3.11	0.338 (0.204)	30/34
EBM-tuned	2.73	0.343 (0.208)	39/25
IGANN-tuned	2.94	0.344 (0.215)	—/—

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5

Model performances for stratified classification datasets in terms of ranking, log loss (LLoss), and wins/losses against IGANN. Standard deviation in parentheses.

Model	Ranking	LLoss	W/L
(a) Small datasets			
Lasso-tuned	4.25*	0.403 (0.184)	6/26
Ridge-tuned	3.31	0.394 (0.182)	15/17
Decision-Tree-tuned	4.97***	0.709 (0.780)	8/24
GAM-Splines-tuned	2.38	0.379 (0.179)	21/11
EBM-tuned	2.94	0.400 (0.182)	19/13
IGANN-tuned	3.16	0.398 (0.196)	—/—
(b) Large datasets			
Lasso-tuned	4.06**	0.342 (0.205)	8/24
Ridge-tuned	3.69*	0.342 (0.204)	8/24
Decision-Tree-tuned	4.16**	0.326 (0.217)	10/22
GAM-Splines-tuned	3.84**	0.296 (0.222)	9/23
EBM-tuned	2.53	0.286 (0.220)	20/12
IGANN-tuned	2.72	0.290 (0.222)	—/—
(c) Low-dimensional			
Lasso-tuned	3.91**	0.445 (0.194)	7/25
Ridge-tuned	3.31	0.441 (0.192)	11/21
Decision-Tree-tuned	4.28***	0.668 (0.793)	11/21
GAM-Splines-tuned	3.56	0.419 (0.205)	10/22
EBM-tuned	3.28	0.443 (0.195)	14/18
IGANN-tuned	2.66	0.421 (0.216)	—/—
(d) High-dimensional			
Lasso-tuned	4.41**	0.300 (0.171)	7/25
Ridge-tuned	3.69	0.295 (0.168)	12/20
Decision-Tree-tuned	4.84***	0.367 (0.237)	7/25
GAM-Splines-tuned	2.66	0.257 (0.171)	20/12
EBM-tuned	2.19	0.243 (0.171)	25/7
IGANN-tuned	3.22	0.267 (0.187)	—/—

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

achieves the second-best rank, outperforming three out of five baseline models on the 0.05 (Lasso, Decision-Tree, GAM-Splines) and one on the 0.1 (Ridge) significance level.

5. Real-world application of IGANN

In this section, we showcase the value of IGANN based on three real-world applications for productivity prediction (Section 5.1), credit scoring (Section 5.2), and criminal recidivism prediction (Section 5.3).

5.1. Case study I: productivity prediction

In the first case, we use an illustrative regression scenario to demonstrate IGANN's intrinsic model interpretability. For this purpose, we chose a productivity prediction case from a garment manufacturing unit in Bangladesh (Imran et al., 2019). The manufac-

turing unit collected production data from twelve teams across two departments for three months, comprising twelve features and 1197 samples. The goal is to predict the unit's productivity with values in the training set ranging from 0.23 (low productivity) to 1.12 (high productivity). Further details about the dataset can be found in Appendix H. For our demonstration, we removed features with limited predictive power, applied common preprocessing steps (e.g., one-hot encoding for categorical features, standardization for continuous features), and fitted IGANN for the regression task.⁶

As shown in Fig. 4, IGANN yields functional relationships to characterize the effect of input feature values on the target value, which are captured by the corresponding shape functions $S_k(x_k)$. The x-axes represent the feature values and the y-axes represent the impact on the target variable. The gray dashed line at $y = 0$ indicates the reference line for the baseline level of each feature. For one-hot encoded categorical features, for instance, this line corresponds to the respective reference group (in this example $Team = 1$ and $Department = Finishing$). The prediction value at the baseline equals the intercept b (in this example $b = 0.7334$). Feature effects above the reference line have a positive impact and feature effects below the reference line have a negative impact on the predicted output.⁷ For example, a targeted productivity of 0.35 decreases the prediction by 0.1824 compared to the reference value when all other features remain constant.

Based on the above, the shape functions directly visualize how a feature affects the prediction target over its entire range of values, offering *intrinsic global interpretability*. Likewise, we can calculate the prediction score for any sample based on Eq. (14), offering *intrinsic local interpretability*. For instance, using the sample highlighted in the gray boxes in Fig. 4 with the respective feature values, we obtain the following prediction: $0.6192 = \hat{y} = b + S_1(Team = 4) + S_2(TargetedProductivity = 0.35) + S_3(Incentive = 0) + S_4(StandardMinuteValue = 4.3) + S_5(NoOfWorkers = 20) + S_6(WorkInProgress = 1190.466) + S_7(OverTime = 3600) + S_7(Department = Finishing) = 0.7334 + 0.047 - 0.1824 - 0.0588 + 0.1151 - 0.0443 + 0.0041 + 0.005 + 0$. The example shows that the shape plots are not just an explanation of relevant relationships, but an exact description of how IGANN computes a prediction. Thus, our model provides a transparent decision logic for comprehensible decision support.

5.2. Case study II: credit default prediction

In the second example, we consider a classification task from the financial sector. Specifically, we use a credit scoring case from the United States where the credit industry is highly regulated by federal laws, such as the Fair Credit Reporting Act (FCRA) and the Equal Credit Opportunity Act (ECOA). That is, financial institutions are legally obliged to provide explanations when responding to customer inquiries or rejecting a loan to ensure a fair and transparent lending process (Chen et al., 2022). The specific case is selected from a large data science competition which was collaboratively organized between the Fair Isaac Corporation, Google, and academics at Berkeley, Oxford, Imperial, UC Irvine, and MIT (FICO, 2018). It focuses on an anonymized dataset of 10,459 real homeowners having requested a home equity credit line. The dataset contains 23 features related to the individuals' credit history and a target variable stating whether or not a homeowner defaulted on a loan. A full description of all features, the applied preprocessing

⁶ See Appendix H for further details about preprocessing, training, and evaluation procedures.

⁷ Note at that point that our conclusions are drawn based on correlations and can not be interpreted as causal effects.

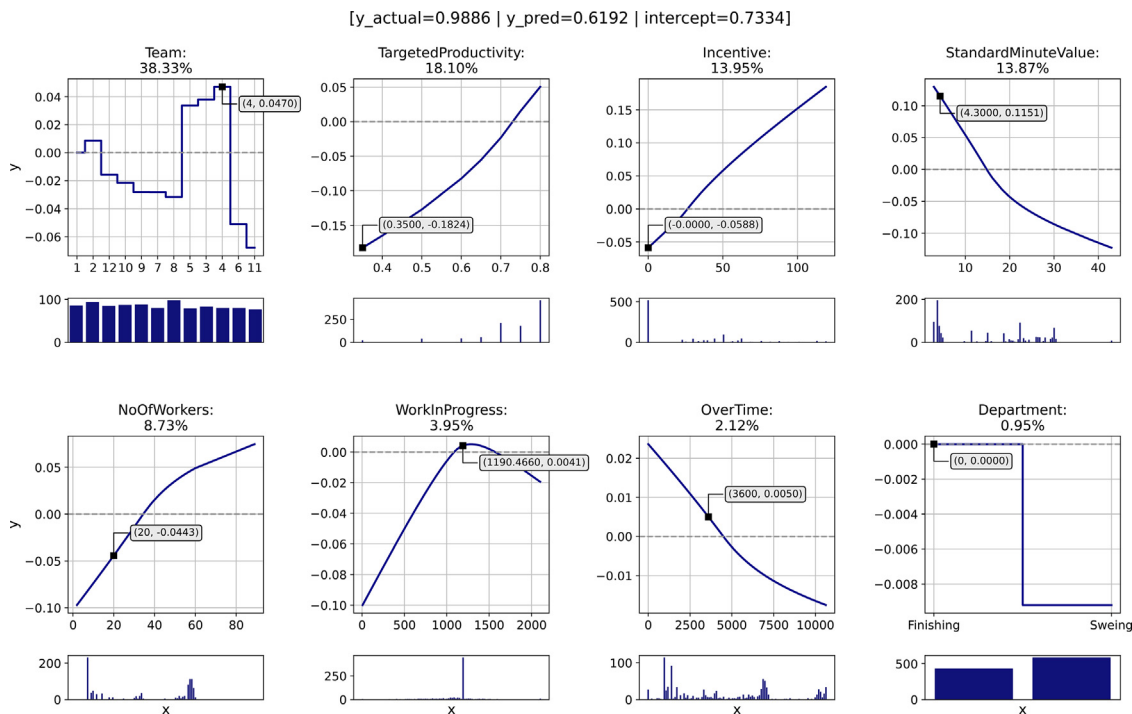


Fig. 4. Results of IGANN for the Productivity dataset based on a subset of eight features. The line plots visualize the effects of input feature values on the target value. The subjunct plots illustrate the distribution of the feature values. The percentages in the subtitles denote the feature contribution. The gray boxes reveal the individual feature effect of a specific sample for local interpretation purposes. The predicted productivity $\hat{y}$ of the selected sample is 0.6192.

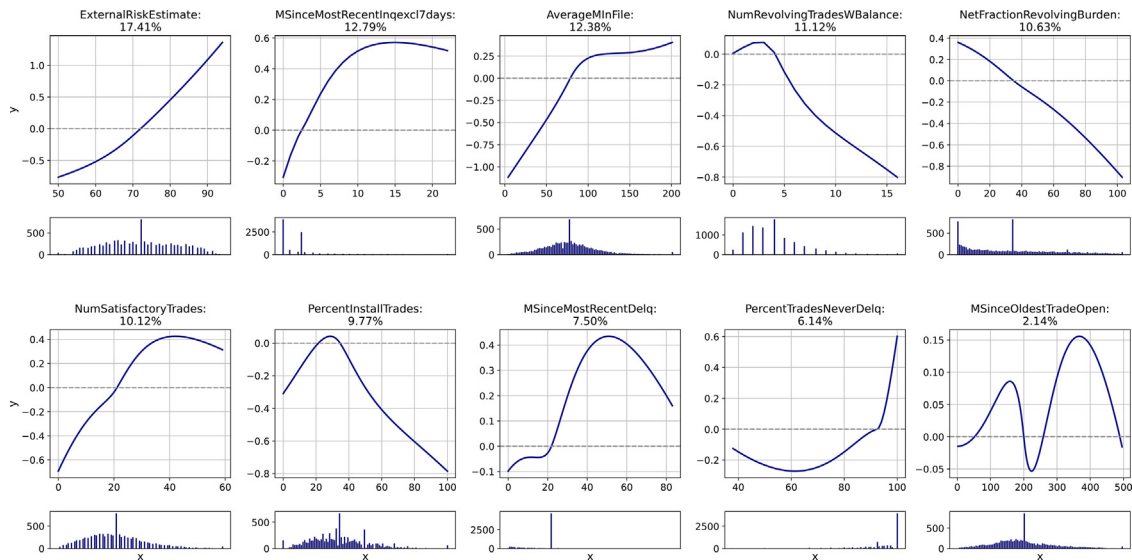


Fig. 5. Results of IGANN for the FICO dataset based on a subset of ten features. The line plots visualize the effects of input feature values on the target value. The subjunct plots illustrate the distribution of the feature values. The percentages in the subtitles denote the feature contribution.

steps, and further details on model training and evaluation can be found in Appendix I.

Fig. 5 visualizes the shape functions of IGANN based on a subset of ten features. As in the first case, our model provides transparent insights about the captured relationships.⁸ For example, it is observable that the feature *MSinceMostRecentDelq* (= months since most recent delinquency) has an overall non-linear effect in logit space with a small negative impact on the prediction tar-

get for feature values smaller than 22 months. For larger values, the effect steadily increases until a feature value of about 50 months. After this point, the impact eventually decreases again. Furthermore, it can be observed that several shape functions have a fairly linear dependency in logit space which either applies to the entire shape plot (e.g., *ExternalRiskEstimate*, *NetFractionRevolvingBurden*) or to certain feature ranges (e.g., *AverageMinFile* on [0, 100], *PercentInstallTrades* on [30, 100]). This allows them to be easily interpreted and communicated for the specific task of credit scoring as the domain is usually dominated by linear scoring models.

⁸ Please note that the y-axes in the shape plots represent the logit space for binary classification tasks. That is, to obtain the final prediction probabilities the sum of the logits need to be transformed using the logistic function.

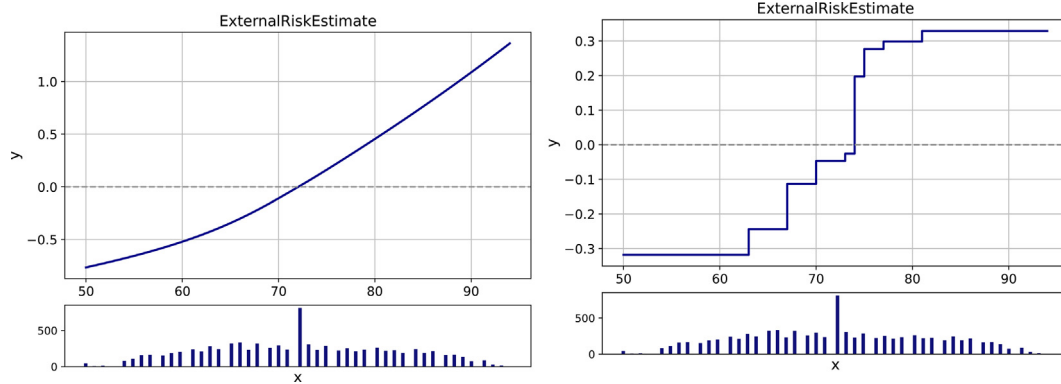


Fig. 6. Shape plot of IGANN (left) and EBM (right) for the feature *ExternalRiskEstimate*.

At this point, IGANN shows its superiority over other interpretable models like EBM and logistic regression. That is, on the one hand, simple relationships can be captured in their natural form, making them more accessible and understandable than erratic and possibly misleading shape functions, such as those fitted by EBM (cf. Fig. 6). On the other hand, IGANN is able to deviate from purely linear relationships whenever the underlying data requires it. This allows IGANN to achieve a sufficient predictive performance, which in the particular case is even better than the performance of flexible black box models like XGBoost and deep neural networks, or of alternative interpretable models like EBM and GAMI-Net. Appendix I provides supplementary evaluation results for the specific dataset by comparing IGANN to a variety of baseline models.

5.3. Case study III: criminal recidivism prediction

In the third case, we consider the highly sensitive task of predicting criminal recidivism (Wang et al., 2023). The purpose of predicting recidivism in the criminal justice system is to assess the likelihood that a criminal defendant will commit another criminal offense in the future. These predictions are used during the pretrial, parole, and sentencing process. In such a high-stakes decision, full model transparency is paramount to avoid discriminatory effects and to ensure algorithmic fairness. Specifically, we consider COMPAS (Correctional Offender Management Profiling for Alternative Sanctions), which is a proprietary recidivism scoring software used by the U.S. courts that was claimed to produce racially-biased predictions. The claim was made by ProPublica through a publicly released data analysis, containing the criminal history and court data of more than 10,000 defendants from Broward County, Florida (Angwin et al., 2016). For demonstration purposes, we use a pre-filtered subset of the data (Agarwal et al., 2021) and show how IGANN is able to detect and mitigate discriminatory biases. The dataset consists of 6172 samples, three demographic features (i.e., *Race*, *Age*, *Sex*), three criminal history features (i.e., *LengthOfStay*, *PriorCount* (count of prior criminal records), *CChargeDegree* (degree of criminal charge)), and a binary target variable stating whether or not a defendant has recidivated within a time frame of two years.⁹

After fitting the model on all six features, IGANN provides corresponding shape plots⁸ for model inspection purposes, see Fig. 7. For example, it can be seen that an increasing count of prior criminal records leads to an increasing risk of recidivism, with a flattening slope starting at a feature value of 5. Age, on the other

hand, reveals the opposite effect, with younger defendants generally showing a higher tendency to recidivate. Apart from that, the shape plots also reveal several discrimination effects originating from potentially biased training data. For instance, compared to the reference group of Native Americans, African-American defendants receive higher prediction scores for recidivism than defendants from other backgrounds.

To address such a racial bias, IGANN can be easily modified by taking advantage of its additive model structure. Specifically, we can remove the learned feature effect *after* model training by setting the shape function of each race category to zero. In other words, all racial groups receive the same prediction score as the reference group of Native Americans, when all other features are held constant (i.e., $S(\text{Race}) = 0$). Depending on what is considered fair, it is also conceivable to set the shape function value of each race category to the weighted average feature effect across all six groups (i.e., $S(\text{Race}) = 0.329$), represented by the red line in Fig. 7. We provide additional insights regarding these different choices in Appendix K. In either case, however, any post-hoc modifications can be incorporated into the model with minimal effort to ensure that defendants of different racial backgrounds are treated more equally. If, by contrast, the offending feature is removed *prior* to model training, debiasing may become even more challenging because the risk arises that the racial bias will spread to proxy variables. That is, instead of eliminating the discrimination effect, it will transfer to other correlated features and thereby remains unexplored (Agarwal et al., 2021). Therefore, IGANN offers a useful way for detecting and mitigating unwanted discrimination effects to ensure a fair decision model.

6. Discussion

6.1. Implications for research

Machine learning models have become an indispensable anchor for proactive decision support tasks that require high prediction qualities. However, in high-stakes decision scenarios such as healthcare, justice, and finance, it is not enough to focus solely on accurate predictions by gradually increasing the model complexity. Instead, it is necessary to obtain powerful yet transparent models that allow for a complete verification of the internal decision logic. In particular, knowledge of the learned functioning of a machine learning model is crucial for several reasons. First, developers and researchers can improve the data preprocessing steps and model decisions. Second, the potential issues that can arise from changing input distributions, i.e., distribution shifts, can be detected. Finally, an awareness of the unwanted biases in the underlying datasets can be raised and addressed accordingly.

⁹ In Appendix J, we provide more details on the features, applied preprocessing steps, model training, and model evaluation.

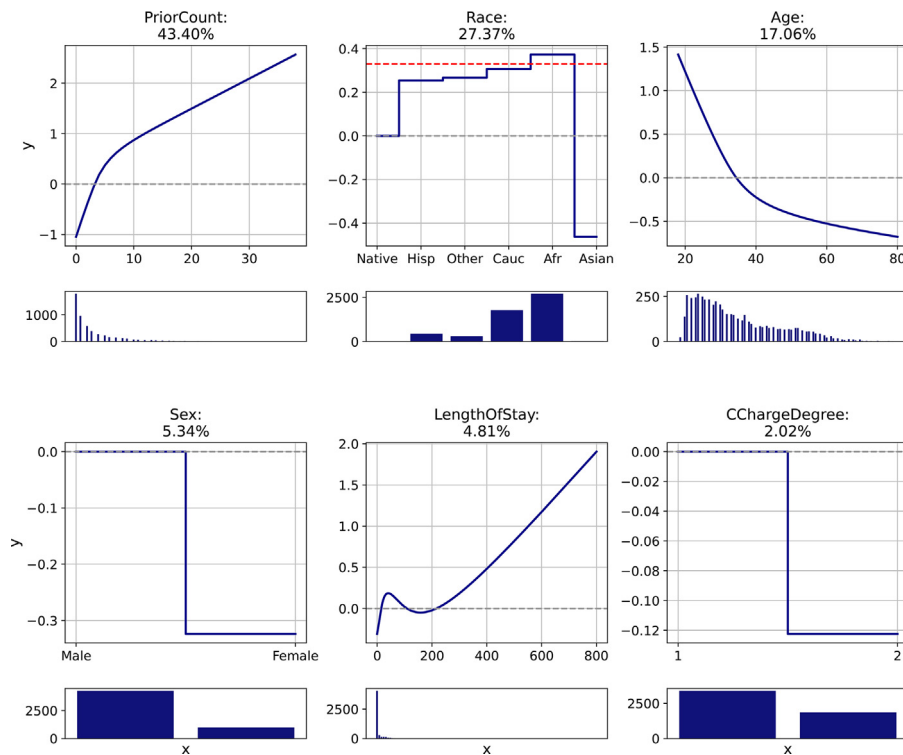


Fig. 7. Results of IGANN for the COMPAS dataset. The line plots visualize the effects of input feature values on the target value. The subplots illustrate the distribution of the feature values. The percentages in the subtitles denote the feature contribution.

One model family that meets both these pivotal objectives (i.e., strong predictive performance and full model transparency) is that of generalized additive models, which have recently gained increasing attention in the research community (e.g., [Lou et al., 2012](#); [Yang et al., 2021b](#)). IGANN complements these developments by introducing a novel and innovative way of model fitting. By proposing this model, we make a manifold of design choices to bolster the predictive performance and obtain a strong off-the-shelf machine learning model for daily use in data analytics.

First, IGANN initializes the shape functions linearly and then gradually adapts to potential non-linearities. It is, therefore, a careful approach that IGANN pursues to keep model complexity low. The second innovation is the combination of gradient boosting with the theory of extreme learning machines. The latter allows to establish a linear time complexity that results in fast training, even for datasets that comprise hundreds of thousands of samples. Furthermore, criticized aspects of the theory of extreme learning machines are naturally mitigated and turned into a strength by our combination with a gradient boosting framework. More specifically, extreme learning machines have been criticized for not being sufficiently reliable due to the dependence on random input weights ([Neumann & Steil, 2011](#)). In our framework, however, this property turns into an advantage when combined with the idea of gradient boosting, as weak learners are supposed to have a random component. To the best of our knowledge, this is the first time that these two fundamental machine learning principles have been combined within a single architecture and it could be demonstrated that the resulting model not only shows an extremely fast learning behavior but also offers excellent prediction results.

Despite the strong predictive performance, it can be pointed out that there is no single best-performing machine learning model across all datasets. On some datasets, linear models outperform complex non-linear models due to their simplicity and robustness. On other datasets, non-linear models are key to obtaining good

predictive performances. Within non-linear models, there can also be no clear winner across more than one hundred tasks. Nevertheless, we observe that, on average, IGANN provides the best performance on a representative sample of 50 regression tasks, and comes second across 64 classification tasks.

6.2. Practical considerations

There are several benefits to using interpretable models in practice. For one, businesses can get a better understanding of how machine learning models work and what factors they take into account when making predictions. This is important for ensuring that a model is accurate and for interpreting its results. Additionally, an interpretable model can help businesses identifying areas where they can improve their processes. For example, if the model is trained on customer data, it may be able to identify patterns in customer behavior that the business can use to improve its marketing or sales strategies. Or if the model is trained on manufacturing data, it may be able to identify process improvements that can be made to increase efficiency.

Consequently, traditional interpretable machine learning models, such as linear models, decision trees, decision rules, or combinations thereof are still used extensively in practice (e.g., [De Caigny et al., 2018](#); [Dumitrescu et al., 2022](#)). Modern GAM approaches complement these developments as they provide an intrinsically interpretable model structure that ensures a high degree of transparency while offering sufficient flexibility to capture more complex relationships. However, although several innovative GAM alternatives have been proposed in the past, only a limited number of algorithms come with a simply applicable implementation.

To this end, we pursued the goal of developing a model that is characterized by its simple applicability. Specifically, we proposed a generalized additive model that is built on shallow neural networks and comes with an efficient training algorithm. IGANN has a linear time complexity and is implemented using PyTorch,

which allows the model to scale to huge datasets. The purpose of IGANN is to serve the community of data scientists as a valuable tool. Therefore, the source code is publicly available under an open source license so that other machine learning researchers and developers can reuse this approach for benchmarking purposes or to extend it with further innovations.

6.3. Limitations

IGANN shows promising results in achieving high predictive performance while maintaining intrinsic interpretability. Yet, like any other machine learning method, it is not without limitations.

First, different stakeholders may have varying expectations and requirements for interpretability depending on the specific use case. Therefore, some applications may require even more restrictive model constraints to satisfy specific needs. For example, Chen et al. (2022) emphasize the requirement for monotonicity constraints in credit scoring models. This shall ensure that an increase in risk factors also increases the predicted risk, which is beneficial for translating model outputs into simple scoring tables. Another requirement could be the construction of sparse models to avoid an unnecessarily large number of features, which was highlighted by Wang et al. (2023) in the context of recidivism predictions. Such additional constraints are not explicitly addressed in the current version of our proposal. However, due to IGANN's flexible model structure, they can be incorporated into future extensions.

Second, IGANN currently assumes no interactions between input features. This restriction is mandatory to ensure IGANN's intrinsic interpretability so that its entire reasoning process is reflected in the computation of feature-wise shape functions that can be easily visualized in a human-understandable manner. At the same time, however, this restriction may also impair predictive performance due to suppressed interaction effects, which could also be observed in our case studies when comparing our model with more flexible black box approaches like XGBoost. As a compromise, we plan to extend IGANN's model structure in future work by incorporating second-order interaction effects, which can still be visualized as human-understandable heatmaps on two-dimensional planes showing the joint effect of two combined features (Lou et al., 2013). For higher-order interactions, on the other hand, or when the interaction structure is too complex, careful feature engineering approaches and additional modeling techniques may be required to uncover underlying interaction effects.

Finally, IGANN can suffer from similar problems as other machine learning models. For example, IGANN's predictive performance may suffer from unbalanced datasets, and its interpretability may become unclear for datasets with high multicollinearity or numerous features. To address these issues, it is necessary to employ strategies such as oversampling or undersampling for unbalanced data and incorporate feature selection techniques for datasets with high multicollinearity or many features.

7. Conclusion

This work proposed IGANN, an innovative intrinsically interpretable machine learning model. Specifically, we propose a new model architecture that makes use of sparsified single hidden layer neural networks to assure full model transparency. We derive an efficient training algorithm based on the theory of extreme learning machines, that allows reducing the training process to solving a sequence of regularized linear regressions. Extensive computational experiments demonstrated that IGANN represents a strong off-the-shelf machine learning model for the daily use in data analytics. We showcase the practical usage on three case studies from the fields of manufacturing, credit default prediction, and criminal

recidivism prediction. In all case studies, robust and comprehensible outcomes are obtained.

The aforementioned method opens avenues for future research. One promising direction is to extend IGANN by incorporating second-order interaction effects to further increase model flexibility and predictive performance. Moreover, one could investigate semi-continuous shape functions with a predefined number of discontinuities. This could be interesting in scenarios, where features have a general smooth effect on the model outcome, but also have narrow feature ranges in which sharp changes appear (e.g., a feature that represents age and the point in time when a person retires). Other interesting research directions could include additional structural constraints on the shape functions, such as monotonicity constraints, or upper and lower bounds. Here, IGANN as a versatile generalized additive model provides a solid starting point for such extensions.

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Supplementary material

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